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Avengers Assembly: We Didn't Know What You'd Like, So We Put Everything In Ramses

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Abstract. We present the non-standard cosmology suite of the cuRAMSES adaptive mesh refinement (AMR) code (cuRAMSES III), which assembles, within a single code base, essentially every popular extension of ΛCDM: (i) **self-interacting dark matter** (SIDM) with constant, velocity-dependent, anisotropic and inelastic (two-state) cross-sections, implemented as cell-based Monte Carlo scattering with machine-precision momentum and energy conservation; (ii) a **dark energy and non-GR gravity** battery comprising CPL dark energy with clustering perturbations, massive-neutrino linear response, early dark energy, field-level quintessence and purely kinetic k-essence, coupled quintessence, parametrized quasi-static Horndeski gravity, Hu–Sawicki $f(R)$, nDGP, symmetron, the environmentally damped dilaton, the parameter-free tracker cubic Galileon, and QUMOND — all built on a unified Poisson-solver framework in which model corrections compose multiplicatively, with an FFT-accelerated Newton stage for the nonlinear scalar fields; (iii) **fuzzy dark matter** (FDM), a hybrid Schrödinger–Poisson integrator with a spectral kinetic operator on the base level and finite differences on refined levels; and (iv) **evaporating primordial black holes** (PBH), a table-driven global mass-function scheme with optional local heating. Every dark-energy and gravity module has been verified at the equation level against the literature, at the solver level against analytic linear responses (to 10^{-5} or better), and end-to-end with a controlled 64^3 simulation suite compared against an initial-conditions-consistent linear forward model, agreeing to 2–5 per cent with residual signs explained by screening. We document the algorithms, unit conventions, namelist interfaces and the verification methodology, providing a self-contained reference for non-standard cosmological simulations with AMR codes.

Keywords: cosmological simulations, dark energy theory, modified gravity, dark matter theory

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Contents

1 Introduction

The ΛCDM concordance model provides an excellent fit to a wide range of cosmological observations [34], yet the physical nature of both dark components remains unknown. On the dark-energy side, the cosmological constant may be replaced by a dynamical field with an evolving equation of state [10, 26], by an early dark energy component [35], or the acceleration may signal a breakdown of General Relativity on cosmological scales, as in $f(R)$ gravity [16], braneworld models [41], Galileon [4], symmetron and dilaton theories [5, 14], or MOND [31]. On the dark-matter side, small-scale tensions motivate self-interacting dark matter [42, 45], ultralight (fuzzy) dark matter [17, 40], and primordial black holes [8], while neutrino oscillations guarantee a small but non-zero hot component [22].

Stage-IV surveys will constrain all of these at the per-cent level, and their nonlinear signatures can only be predicted by simulations. Historically each extension has lived in its own specialised code; comparing models then convolves physical differences with code systematics. This paper describes the opposite approach: one AMR code, cuRAMSES [20], carrying *all* of the above extensions behind uniform namelist switches, with shared infrastructure (domain decomposition, Morton-key octree, multigrid/CG/FFT Poisson solvers, OpenMP+MPI parallelism) and a common verification methodology. Hence the title[1].

The paper is organised in four physics sections, one per family: Section 2 describes self-interacting dark matter; Section 3 the dark energy and non-GR gravity battery, including the unified Poisson framework, the nonlinear scalar-field solvers and their verification; Section 4 fuzzy dark matter; and Section 5 evaporating primordial black holes. Section 6 concludes. Throughout, code quantities use the supercomoving variables of Martel & Shapiro [29]: $\tilde{x} = x_{\rm com}/(aB)$ (with B the box size), $\tilde{\rho} = \rho a^3/(\rho_c \Omega_m)$ (so $\bar{\tilde{\rho}} = 1$), $\tilde{\phi} = a^2\phi/(BH_0)^2$ and $\mathrm{d}\tilde{t} = H_0\,\mathrm{d}t/a^2$.

2 Self-Interacting Dark Matter

2.1 Physical motivation

SIDM models introduce a non-zero scattering cross-section σ/m that allows dark matter particles to exchange momentum and energy, potentially resolving the core–cusp, missing satellites and too-big-to-fail problems [42, 45]. Astrophysical constraints span dwarf galaxies ($v \sim 30$–$100\,\mathrm{km\,s^{-1}}$) to clusters ($v \sim 10^3$–$3\times10^3\,\mathrm{km\,s^{-1}}$); a constant $\sigma/m \sim 1\,\mathrm{cm^2\,g^{-1}}$ fits dwarf scales but may violate cluster bounds, motivating velocity-dependent cross-sections from light-mediator models [12, 19, 46].

2.2 Monte Carlo scattering algorithm

We implement cell-based Monte Carlo scattering following the standard mesh/SPH approach [37, 38]. In every AMR leaf cell, dark matter particles are collected from the particle linked list, shuffled with the Fisher–Yates algorithm, and paired sequentially. For a pair with relative velocity $v_{\rm rel}$ the scattering probability per step is

$$P \;=\; \frac{\sigma/m(v)\,\bar{m}\,v_{\rm rel}\,\Delta t}{V_{\rm cell}}\,(N_{\rm dm}-1), \tag{2.1}$$

with $\bar{m}$ the mean particle mass, $V_{\rm cell}$ the proper cell volume and the factor $(N_{\rm dm}-1)$ accounting for all possible partners; a further factor $N_{\rm dm}/(2\lfloor N_{\rm dm}/2\rfloor)$ makes the $\lfloor N_{\rm dm}/2\rfloor$ sampled pairs reproduce the exact $N_{\rm dm}(N_{\rm dm}-1)/2$ pair rate for odd occupancies. A scatter occurs when a uniform deviate falls below P; post-scattering velocities are constructed in the

[1] From the Korean "*mwol joahalji mollaseo da neoeosseo*" — we didn't know what you'd like, so we put everything in.

centre-of-mass frame,

$$\boldsymbol{v}_i' = \boldsymbol{v}_{\rm cm} + \frac{m_j}{m_i+m_j}\,|v_{\rm rel}'|\,\hat{\boldsymbol{n}}, \qquad \boldsymbol{v}_j' = \boldsymbol{v}_{\rm cm} - \frac{m_i}{m_i+m_j}\,|v_{\rm rel}'|\,\hat{\boldsymbol{n}}, \tag{2.2}$$

which conserves momentum by construction and kinetic energy exactly for elastic scattering.

2.3 Cross-sections and angular distributions

Three velocity dependences are supported (`sidm_type`): constant $\sigma/m = \sigma_0$; Yukawa (Born-regime) $\sigma(v)/m = \sigma_0[1 + (v/v_0)^2]^{-2}$; and a generic power law $\sigma(v)/m = \sigma_0(v/v_0)^n$. The angular distribution is isotropic by default, or forward-peaked Rutherford-like, $\mathrm{d}\sigma/\mathrm{d}\Omega \propto (1 - \cos\theta + 2\varepsilon)^{-2}$ [18], sampled by CDF inversion.

2.4 Inelastic (two-state) SIDM

A two-state model [44, 47] adds a mass splitting δ. Because an N-body macro-particle scatters as if a single dark matter particle of mass m_χ (`sidm_mchi`) does, the kinematics are those of the *microphysical* pair: ground–ground pairs up-scatter ($g + g \to e + e$) when $\frac{1}{2}\mu_\chi v_{\rm rel}^2 > 2\delta$ with $\mu_\chi = m_\chi/2$, excited pairs down-scatter exothermically ($v'^2 = v^2 + 8\delta/m_\chi$), and mixed pairs scatter elastically; when an inelastic channel is open it competes with the elastic one with equal weight. Each transition changes the macro-particle kinetic energy by $2\delta\,(m_p/m_\chi)$, which we verify at the per-event level in the test below. An initial excited fraction is assigned at start-up in a restart-safe manner. A dissipative channel (`sidm_fdiss`) and a DM–baryon drag term (`sidm_baryon`) are also available.

2.5 Timestep, threading and diagnostics

The maximum per-level probability $P_{\max}$ is fed back into the timestep as $\Delta t_{\rm new} \le (C_{\rm SIDM}/P_{\max})\,\Delta t_{\rm old}$ (default $C_{\rm SIDM} = 0.1$). The sweep is OpenMP-parallel over grids with per-thread decorrelated RNG streams and deterministic reseeding, and cumulative momentum/energy residuals are reduced across MPI ranks each step, providing runtime verification of the conservation properties. All parameters live in `&SIDM_PARAMS`.

2.6 Validation

On a 64^3-particle unigrid ΛCDM box ($L = 100\,h^{-1}\,$Mpc, $\sigma/m = 10^3\,\mathrm{cm^2\,g^{-1}}$) we compare the per-step scattering counts against the exact Monte-Carlo expectation evaluated offline from the snapshot (cell-by-cell pair sums of $|\Delta\boldsymbol{v}|$): the measured/predicted event ratio is 1.07 ± 0.06 at $a = 0.3$ and 1.006 ± 0.044 at $a = 0.6$, and the per-cell pair counts match exactly, confirming both the rate normalisation and its scale-factor dependence. Momentum and elastic kinetic energy are conserved to machine precision ($|\Delta p|/p \sim 10^{-14}$ per event). For the two-state model with $\delta = 10\,$keV, $m_\chi = 1\,$GeV the up-scattering threshold gates correctly (no transitions in a cold box), a 50 per cent excited population decays through down-scattering only, and the measured per-event energy release matches $2\delta\,m_p/m_\chi$ to four significant figures.

3 Dark Energy and non-GR Gravity

This section describes the largest part of the suite: eleven dark-energy and modified-gravity models sharing one solver infrastructure. Table 1 summarises the models, their parameters and their verification status.

Table 1. The dark energy and gravity battery. "Linear response" lists the analytically validated unscreened limit; the last column the accuracy achieved in dedicated unit tests of the discretised equations (see Section 3.10).

Model	Switch / namelist	Background	Linear response	Unit-test accuracy
CPL DE + perturbations	`w0,wa,cs2_de,de_table`	$w = w_0 + w_a(1-a)$	Sapone–Kunz closure	≤ 0.1–0.8% vs CAMB ($z \geq 0.5$)
Massive ν linear response	`use_neutrino`	—	$1 + (\Omega_\nu/\Omega_{cb})R_\nu(k,a)$	table-exact
Early dark energy	`use_ede`	Poulin+19 fluid	smooth	analytic dilution exact
Quintessence	`use_quintessence`	Klein–Gordon (shooting)	smooth ($c_s^2 = 1$)	w-consistency 2×10^{-5}
k-essence	`use_kessence`	$P(X) = -X + X^2$ (exact)	Helmholtz with $c_s^2(a)$	analytic limits exact
Coupled quintessence	`use_coupled_de`	KG + mass evolution	$G_{\rm eff} = G(1+2\beta^2)$	isolation runs $\leq 1\%$
Horndeski $\mu(a,k)$	`use_horndeski`	ΛCDM/CPL	$\mu = 1 + \mu_0\Omega_{\rm DE}(a)/\Omega_{{\rm DE},0}$	exact by construction
$f(R)$ Hu–Sawicki	`use_fR`	ΛCDM	$\mu = 1 + \frac{1}{3}\frac{k^2}{k^2+m^2a^2}$	6×10^{-7}
nDGP	`use_nDGP`	ΛCDM	$G_{\rm eff}/G = 1 + 1/(3\beta)$	force field 1.5×10^{-6}
Symmetron	`use_symmetron`	ΛCDM	$F_5/F_N = 2\beta^2\bar{\chi}^2$	exact (Brax+12 eq. 68 match)
Dilaton (Brax+12)	`use_dilaton`	ΛCDM	$2\beta(a)^2k^2/(k^2+m^2a^2)$	5 digits
Cubic Galileon (tracker)	`use_galileon`	tracker $E^2(a)$	$G_{\rm eff}/G(a{=}1) = 1.844$	5 digits
QUMOND	`use_mond`	—	$\nu(\lvert g_N\rvert/a_0)$	$\mu\nu$ conjugacy 2×10^{-16}

3.1 Unified Poisson-solver framework

In supercomoving units the code solves $\tilde{\nabla}^2\tilde{\phi} = \frac{3}{2}\Omega_m a(\tilde{\rho} - 1)$. All smooth-dark-sector corrections enter this equation as multiplicative factors on the source. A single function, `cosmo_poisson_fourpi()`, provides the prefactor consistently to the multigrid, conjugate-gradient and coarse-level solvers, and the FFT solvers apply the same corrections per Fourier mode in the Green's function, so that scale-dependent responses cost nothing extra. The factors compose:

$$\frac{3}{2}\Omega_{cb}\, a \;\times\; \underbrace{\left[1 + \frac{\Omega_\nu}{\Omega_{cb}} R_\nu(k,a)\right]}_{\text{neutrinos}} \underbrace{\left[1 + \frac{\Omega_{\rm DE}(a)}{\Omega_{cb}} R_{\rm DE}(k,a)\right]}_{\text{DE clustering}} \underbrace{\mu(a,k)}_{\text{Horndeski}} \underbrace{e^{-\beta(\varphi-\varphi_0)}}_{\text{coupled DE}}. \tag{3.1}$$

3.2 CPL dark energy with clustering perturbations

The background uses the exact CPL integral $f_{\rm DE}(a) = a^{-3(1+w_0+w_a)}e^{-3w_a(1-a)}$ in the supercomoving Friedmann factors, the growth factor and the initial velocity scaling. Perturbations are supported in two modes:

(i) Table mode. A CAMB-derived table of the effective ratio $R_{\rm DE}(k,a) = \delta_{\rm DE}/\delta_m$ is measured with the Weyl-potential method (the ratio of potential per unit matter contrast between a clustering-DE and a smooth-DE universe), which yields the Newtonian-gauge response appropriate for the Poisson source; synchronous-gauge $\delta_{\rm DE}$ would over-correct by an order of magnitude.

(ii) Quasi-static fallback. When no table is supplied and $c_s^2 > 0$, the code applies the Sapone & Kunz [39] closure

$$\frac{\delta_{\rm DE}}{\delta_m} = \frac{1+w}{(1-3w) + \dfrac{2c_s^2k^2}{3a^2H^2}}, \qquad \text{i.e. d.f.} = 1 + \frac{\alpha}{k^2+\kappa^2}, \tag{3.2}$$

with $\kappa^2 = \frac{3}{2}(1-3w)a^2E^2(L/\lambda_H)^2/c_s^2$ and $\alpha = (\Omega_\Lambda f_{\rm DE}a^3/\Omega_m)(1+w)\,\frac{3}{2}a^2E^2(L/\lambda_H)^2/c_s^2$. Correct limits: a k-independent boost $1 + (\rho_{\rm DE}/\rho_m)(1+w)/(1-3w)$ for $c_s^2 \to 0$, unity below the sound horizon, and bit-wise unity for Λ. Validated against CAMB (same Weyl method) to ≤ 0.1 per cent for $(w_0, w_a) = (-0.9, 0.1)$ and ≤ 0.8 per cent for $w_0 = -0.8$ at $z \geq 0.5$ (~ 4 per cent at $z = 0$, where the quasi-static approximation degrades — use the table mode for precision at low redshift). An earlier release used a closure with the opposite sign (a 15–29 per cent source suppression); results obtained with it should be discarded.

3.3 Massive neutrino linear response

Neutrino perturbations are added without neutrino particles: $\delta_\nu(k,a) = R_\nu(k,a)\,\delta_{cb}(k,a)$ with R_ν precomputed from CAMB [23], entering the Poisson source as in Section 3.1. This avoids neutrino shot noise entirely and is exact at linear order [cf. 1].

3.4 Early dark energy

The Poulin et al. [35] fluid form $\rho_{\rm EDE}(a) = \omega_{\rm ede}\,\rho_{\rm std}(a_c)\,2/[1+(a/a_c)^{3(1+w_{\rm ede})}]$ is added to the Friedmann factors. The implementation reproduces the analytic dilution slope $-3(1+w)$ and the analytic peak $f_{\rm EDE}^{\rm max} = 1.14\,\omega_{\rm ede}$ at $a = 3^{1/4}a_c$ exactly. Two conventions matter when mapping CMB constraints: $\omega_{\rm ede}$ is the fraction at a_c (not the peak), and the background carries no radiation, so at $z_c \sim 3000$ the fraction differs from Planck-style $f_{\rm EDE}$ by $\sim 2\times$.

3.5 Scalar-field dark energy: quintessence and k-essence

A dedicated module tabulates scalar-field backgrounds on a $\ln a$ grid ($a = 10^{-6}$–1.55) with lazy, self-healing initialisation (the tables rebuild automatically if the cosmology is overridden by IC headers).

Quintessence [36]: the Klein–Gordon and Friedmann equations are integrated with RK4 for Ratra–Peebles ($V \propto \varphi^{-\alpha}$) or exponential potentials, the amplitude fixed by shooting to $\Omega_\varphi(a{=}1) = \Omega_\Lambda$. The tabulated $w(a)$ obeys $\mathrm{d}\ln f_{\rm DE}/\mathrm{d}\ln a = -3(1+w)$ to 2×10^{-5}, and the RP tracker attractor $w = -2/(\alpha+2)$ is recovered in the matter era.

k-essence [3]: for the purely kinetic $P(X) = -X + X^2$ the shift-symmetry integral $P_{,X}\sqrt{2X}\,a^3 = \text{const}$ gives $X(a)$ algebraically; $w \to 1/3$ and $c_s^2 \to 1/3$ at early times (a dark-radiation-like component) and $w \to -1$ today. The model's $c_s^2(a)$ feeds the Helmholtz closure of equation (3.2) automatically.

3.6 Coupled quintessence

Conformally coupled dark matter, $m_{\rm dm} \propto e^{-\beta\varphi/M_{\rm Pl}}$ [2], adds four effects, all implemented and individually verified: (i) the modified expansion, including the DM mass-evolution factor in the *matter* term of the Friedmann integrals (its omission is detectable: an isolation run reproduced the inconsistent-H growth prediction to 0.2 per cent); (ii) the same factor multiplying the Poisson source; (iii) the momentum-conservation velocity term $+\beta\dot\varphi\,\boldsymbol{v}$, split across the two half-kicks of the KDK integrator; and (iv) the fifth-force enhancement $G_{\rm eff} = G(1+2\beta^2)$ between matter particles.

3.7 Parametrized Horndeski gravity

The quasi-static, scale-independent limit of Horndeski gravity is implemented as $\mu(a) = 1 + \mu_0\,\Omega_{\rm DE}(a)/\Omega_{\rm DE,0}$ multiplying the Poisson source, with an optional Compton-mass k-dependence $\mu(a,k) = 1 + [\mu(a)-1]\,k^2/(k^2 + a^2m^2)$ applied exactly in the FFT solver paths (the multigrid path uses the $k\to\infty$ limit).

3.8 Screened modified gravity: the scalar-field solvers

Five models solve a nonlinear elliptic equation for an extra scalar degree of freedom on the AMR hierarchy, with the fifth force added to the particle acceleration.

3.8.1 $f(R)$ Hu–Sawicki

The quasi-static equation in code units is

$$\tilde{\nabla}^2 f_R = \frac{a^2}{3}\left(\frac{H_0 B}{c}\right)^2\big[\tilde{R}(f_R) - \bar{\tilde{R}}\big] - \Omega_m\left(\frac{H_0 B}{c}\right)^2\frac{\tilde{\rho}-1}{a}, \tag{3.3}$$

with the Hu–Sawicki inversion $\tilde{R} = \bar{\tilde{R}}_0(f_{R0}/f_R)^{1/(n+1)}$ and force $F_5 = +\frac{a^2}{2}(c/H_0B)^2\tilde{\nabla} f_R$ [16, 24, 33]. The linearized mass term is a Yukawa (the Jacobian is negative definite) and the chameleon limit $\tilde{R} \to \bar{\tilde{R}} + 3\Omega_m\delta/a^3$ is exact. The discretised linear response reproduces $\mu(k)-1 = \frac{1}{3}k^2/(k^2+m^2a^2)$ to 6×10^{-7} across $a \in [0.25, 1]$, $f_{R0} \in \{-10^{-5}, -10^{-6}\}$, $n \in \{1, 2\}$.

3.8.2 nDGP

The brane-bending mode obeys the Vainshtein equation [25, 41]; in code units

$$\tilde{\nabla}^2\varphi + \frac{1}{12\,\Omega_{rc}\beta(a)\,a^4}\big[(\tilde{\nabla}^2\varphi)^2 - (\tilde{\nabla}_i\tilde{\nabla}_j\varphi)^2\big] = \frac{\Omega_m a}{\beta(a)}(\tilde{\rho}-1), \tag{3.4}$$

with $\beta = 1 + 2Hr_c(1 + \dot{H}/3H^2)$ and $F_5 = -\frac{1}{2}\tilde{\nabla}\varphi$, so that the unscreened limit is $G_{\rm eff}/G = 1 + 1/(3\beta)$ [49].

3.8.3 Symmetron

The dimensionless field $\chi = \varphi/\varphi_\star$ obeys $\tilde{\nabla}^2\chi = \frac{a^2}{2\tilde{\lambda}^2}[\tilde{\rho}\,(a_{\rm ssb}/a)^3 - 1]\chi + \frac{a^2}{2\tilde{\lambda}^2}\chi^3$ [11, 14], identical to eq. (68) of Brax et al. [6], with force $F_5 = -6\Omega_m\beta^2(\lambda/B)^2(a^2/a_{\rm ssb}^3)\,\chi\tilde{\nabla}\chi$, whose unscreened limit is the canonical $F_5/F_N = 2\beta^2\bar{\chi}^2$. Because $\chi = 0$ is an exact fixed point of the homogeneous equation, the broken-phase VEV $\bar{\chi}(a) = \sqrt{1-(a_{\rm ssb}/a)^3}$ is seeded into uninitialised cells at every solve (this also covers restarts and newly refined cells).

3.8.4 Environmentally damped dilaton

The original $r = 3/2$ dilaton of Brax et al. [5], with $A(\varphi) = 1 + \frac{A_2}{2}\varphi^2/M_{\rm Pl}^2$, $m^2(a) = 3A_2H^2(a)$ and $\beta(a) = \beta_0 a^{3\Omega_m}$ [6], is solved in $\chi = \varphi/M_{\rm Pl}$:

$$\tilde{\nabla}^2\chi = \frac{3\Omega_m A_2 B_2}{a}\,(\tilde{\rho}\,\chi - \bar{\chi}) + a^2 B_2\big[v_\varphi(\chi) - v_\varphi(\bar{\chi})\big], \tag{3.5}$$

with $v_\varphi(\chi) = -3\Omega_m\beta_0(A_2\chi/\beta_0)^{1-3/s}$, $s = 3\Omega_m$, $B_2 = (H_0B/c)^2$ and $F_5 = -(c/H_0B)^2a^2A_2\,\chi\tilde{\nabla}\chi$. The exponent $1-3/s < 0$ makes the Newton Jacobian negative definite. The background is an exact fixed point of the discrete operator, the linear response matches $2\beta(a)^2k^2/(k^2+m^2a^2)$ to five digits, and a $\delta = 10^5$ top-hat drives $\chi \to 0.03\bar{\chi}$ (Damour–Polyakov screening). User parameters are simply β_0 and the present force range $\lambda = 2998\,\xi\,h^{-1}$Mpc (then $A_2 = 1/3\xi^2$).

3.8.5 Tracker cubic Galileon

On the tracker, $H\dot{\varphi} = \xi H_0^2 M_{\rm Pl}$ with $c_2 = -6c_3\xi$, the model is parameter-free: $\xi = \sqrt{6(1-\Omega_m)}$, $E^2(a) = \frac{1}{2}[\Omega_m a^{-3} + \sqrt{\Omega_m^2a^{-6} + 4(1-\Omega_m)}]$ (wired through the background dispatch, so the modified expansion and growth are automatic; $w_0 \simeq -1.18$), and the perturbation coefficients of Barreira et al. [4] reduce to $\beta_1 = \frac{\xi}{3}[2\dot{H}/H^2 - 1 + (1-\Omega_m)/E^4]$, $\beta_2 = 2E^2\beta_1/\xi^2$. The field equation is the nDGP template with coefficient $1/(3\beta_1a^4)$ and source $\Omega_m a\,\delta/\beta_2$; the Poisson back-reaction term integrates to a potential proportional to the field, so the entire fifth force is one gradient, $F_5 = +\frac{\xi}{6E^2}\tilde{\nabla}u$. The unscreened response matches the analytic $-\xi/(9\beta_2E^2)$ to five digits at $a = 0.25$–1: $G_{\rm eff}/G(a{=}1) = 1.844$, decaying as $1/E^4$ into the past. Since $G_{\rm eff}/G - 1 < 10^{-3}$ for $z \gtrsim 2.5$, the solver is skipped at early times.

3.8.6 QUMOND

The quasi-linear formulation of MOND [32] defines a curl-free field via $\nabla^2\Phi = \nabla\cdot[\nu(|\nabla\Phi_N|/a_0)\nabla\Phi_N]$. Both an algebraic force-correction mode and the full phantom-density mode (compute $\rho_{\rm ph} = \nabla\cdot[(\nu-1)\nabla\Phi_N]/4\pi G$, re-solve Poisson) are implemented, with the simple and standard interpolation pairs (verified mutually conjugate to 2×10^{-16}), an external-field option, and a_0 converted to code units through the exact supercomoving acceleration factor (so the MOND scale is correctly constant in proper units at all redshifts). An AQUAL fixed-point mode is also available [cf. 7, 28].

3.9 Numerical solution of the scalar equations

All five scalar solvers share one machinery: nonlinear Newton–Gauss–Seidel relaxation with true three-dimensional red–black colouring (cell parity $i+j+k$), Neumann closure of the stencil at coarse–fine boundaries, one-sided force gradients at level edges, and MPI ghost exchange each sweep. On the fully refined domain level the long-wavelength modes — which plain relaxation cannot converge in $\mathcal{O}(10)$ sweeps — are solved spectrally: a Newton–Helmholtz stage $(\tilde\nabla^2 - \bar m^2)\,\delta u = -\mathcal{R}$ with the level-averaged Jacobian mass for $f(R)$/symmetron/dilaton, and a Winther et al. [49]-style operator splitting for the Vainshtein-type models (solve the cell-local quadratic for $\tilde\nabla^2\varphi$, then invert the Laplacian by FFT). In a controlled test the operator-split stage reached a lower residual in five iterations than 2×10^4 damped relaxation sweeps. Two MPI subtleties deserve mention as a warning to implementers: every rank must enter the per-level ALLREDUCE of the convergence test and the final force ghost-exchange *even when it owns no grids on that level*; early returns on empty ranks deadlock or abort the run at the first sparsely refined level.

3.10 Verification

Verification proceeded on three levels.

Equation level. Every model was re-derived from its original reference (unit conversions included) and the code compared term by term; this stage found and fixed, among others, a wrong-sign DE-perturbation closure, a tachyonic (non-convergent) $f(R)$ operator, a double-counted nDGP coupling, a symmetron force normalisation missing the box-unit conversion, and a zero-pinned symmetron field.

Solver level. The discretised operators were exercised in isolation against analytic linear responses; the accuracies are listed in Table 1.

End-to-end level. A controlled suite of 64^3, $100\,h^{-1}$Mpc dark-matter-only simulations was run from shared Zel'dovich initial conditions ($z=49$, CAMB spectrum) for ΛCDM, quintessence, k-essence, Horndeski, coupled DE (plus per-effect isolation runs), the tracker Galileon and the dilaton. The measured $P_{\rm model}/P_{\Lambda\rm CDM}$ in the band $k = 0.05$–$0.12\,h\,{\rm Mpc}^{-1}$ at $z\simeq0$ was compared against an *initial-conditions-consistent* linear forward model, which (i) rescales the initial displacements by each model's own velocity conversion factor — RAMSES converts the shared velocity files with the model's $v_{\rm fact}$, an effect reaching $(1.077)^2$ for k-essence and $(1.109)^2$ for coupled DE — and (ii) propagates the exact (δ,θ) initial vector through each model's linear system. Results:

All models agree to 2–5 per cent; the Galileon and dilaton deficits carry the expected sign, since Vainshtein and Damour–Polyakov screening remove part of the fifth-force boost already at quasi-linear wavenumbers [cf. 4]. Each coupled-DE ingredient (mass evolution, friction, $G_{\rm eff}$) was additionally isolated with toggle runs and verified at the ~ 1 per cent

Table 2. End-to-end suite: simulated vs. predicted $P/P_{\Lambda\mathrm{CDM}}$ ($k = 0.05$–$0.12\,h/\mathrm{Mpc}$, $z \simeq 0$).

Model	Simulation	Forward model	Ratio
Quintessence (RP α=1)	0.832	0.817	1.018
k-essence (X_0/M^4=0.5005)	0.875	0.881	0.993
Horndeski (μ_0=0.2)	1.036	1.052	0.985
Coupled DE (β=0.1)	0.966	1.005	0.961
Galileon (tracker)	1.148	1.208	0.950
Dilaton (β_0=0.5, λ=20)	1.039	1.071	0.970

level. The forward model, the IC generator and all unit-test drivers ship with the code (`patch/cuRamses/aux/`).

4 Fuzzy Dark Matter

4.1 Physical motivation and equations

Ultralight scalar ("fuzzy") dark matter with $m_a \sim 10^{-23}$–$10^{-21}\,\mathrm{eV}$ [17] behaves as cold dark matter on large scales while suppressing structure below its de Broglie wavelength and producing solitonic halo cores and interference granules [40]. On scales far below the Compton wavelength the field obeys the comoving Schrödinger–Poisson (SP) system [48],

$$\mathrm{i}\hbar\,\partial_t\psi \;=\; -\frac{\hbar^2}{2m_a a^2}\,\nabla^2\psi \;+\; m_a\Phi\,\psi, \qquad \nabla^2\Phi \;=\; 4\pi G a^2\,\bar{\rho}_\mathrm{m}\,\delta, \tag{4.1}$$

with $|\psi|^2$ the comoving dark-matter density and Φ the peculiar potential.

4.2 Supercomoving formulation

RAMSES integrates cosmology in the supercomoving variables of Martel & Shapiro [29]: code time obeys $\mathrm{d}\tilde{t} = H_0\,\mathrm{d}t/a^2$ and positions are comoving box coordinates. A property of this gauge worth stating explicitly — because overlooking it silently freezes structure formation — is that *every explicit scale factor drops out of the SP system.* In code units eq. (4.1) becomes

$$\mathrm{i}\,\partial_{\tilde{t}}\psi \;=\; -\frac{\tilde{\hbar}}{2}\,\tilde{\nabla}^2\psi \;+\; \frac{\tilde{\Phi}}{\tilde{\hbar}}\,\psi, \qquad \tilde{\hbar} \;\equiv\; \frac{\hbar}{m_a L^2 H_0} \;=\; \mathrm{const}, \tag{4.2}$$

where L is the comoving box size: the kinetic $1/a^2$ is absorbed exactly by the supercomoving time, and the expansion enters only through the standard RAMSES Poisson source $\frac{3}{2}\Omega_\mathrm{m} a\,\delta$. The wave sector thereby mirrors the particle sector, where kicks and drifts likewise carry no scale factors. Retaining the naive $1/a^2$ of eq. (4.1) in the supercomoving drift boosts the effective $\tilde{\hbar}$ by a^{-2} ($\sim 10^3$ at $z = 30$), which suppresses gravitational growth at all wavenumbers; we verified that with eq. (4.2) the measured power spectrum growth between snapshots reproduces linear theory (section 4.5).

The integrator is drift–kick–drift per AMR level. On the (uniform) base level the kinetic half-step is applied spectrally, $\hat{\psi}(\boldsymbol{k}) \to \hat{\psi}(\boldsymbol{k})\,e^{\mathrm{i}\tilde{\hbar}\lambda(\boldsymbol{k})\Delta\tilde{t}/2}$, with $\lambda(\boldsymbol{k})$ the discrete 7-point Laplacian eigenvalue, using a distributed in-place complex-to-complex FFT whose forward transform is left in transposed layout to halve the all-to-all volume. On refined levels the kinetic operator is the Crank–Nicolson (Cayley) form $(1 - \mathrm{i}g\mathcal{L})^{-1}(1 + \mathrm{i}g\mathcal{L})$, $g = \tilde{\hbar}\Delta\tilde{t}/4\Delta x^2$

L [h^{-1}Mpc]	valid range (rms $\Delta\theta < \pi$)	conservative ($< \pi/2$)
1	all z	all z
2	$z > 0.7$	$z > 1.8$
4	$z > 3.0$	$z > 5.4$
8	$z > 7.8$	$z > 13$
16	$z > 18$	$z > 29$
≥32	— (IC already aliased)	—

Table 3. Redshift range over which a 1024^3 wave simulation with $m_a = 10^{-22}$ eV resolves the rms bulk-flow phase gradient, as a function of comoving box size. The limit scales as $L_{\max}\sigma_v(L_{\max}) \propto N/m_a$.

— unitary and unconditionally stable — solved matrix-free with a complex BiCGSTAB iteration; a fourth-order Yoshida composition is optional (`fdm_split_order=4`). The potential kick multiplies $e^{-i\tilde{\Phi}\Delta\tilde{t}/\tilde{\hbar}}$ in real space. Initial conditions are constructed by the Madelung map from the same **`grafic`** files used for particle runs: $|\psi| = \sqrt{1+\delta}$ and phase $S/\tilde{\hbar}$ with $\nabla S = \boldsymbol{v}$ obtained by an FFT inverse gradient of the velocity fields, discretely consistent with the particle initialisation.

4.3 Phase resolvability sets the box size

A wave discretisation must resolve the local phase gradient: a flow of peculiar velocity v imprints $\Delta\theta = m_a a v\, \Delta x/\hbar$ radians per cell, and once $\Delta\theta > \pi$ the sampled phase aliases to a wrong velocity — bulk transport, and with it large-scale growth, is corrupted [30]. Because the rms bulk velocity grows with box size while $\Delta x = L/N$ grows at fixed N, this imposes a joint limit on (L, N, m_a, z); the constraint tightens $\propto a^{3/2}$, so late times are the bottleneck. Table 3 lists the validity ranges for $N = 1024^3$ and $m_a = 10^{-22}$ eV computed from the linear velocity power restricted to box modes: reaching $z = 0$ requires $L \lesssim 1$–$2\, h^{-1}$Mpc, in line with the box sizes of published wave simulations, and the reach scales as $L_{\max}\sigma_v(L_{\max}) \propto N/m_a$. A $128\, h^{-1}$Mpc box at 1024^3 carries an rms phase jump of 5.7π per cell already in the $z = 30$ initial conditions — pure wave evolution is excluded there at any redshift, motivating the hybrid scheme below.

4.4 Hybrid fluid–wave scheme

Following the strategy of GAMER-2 [9, 21], the hybrid mode (`fdm_use_hjm`) evolves coarse levels ($\ell <$ `fdm_first_wave_level`) as a Hamilton–Jacobi–Madelung (HJM) fluid and fine levels with the wave solver. On fluid levels the state is stored *persistently* as (ρ, S) in the $(\psi_{\rm re}, \psi_{\rm im})$ arrays, with the action S kept as an *unwrapped* real field: no atan2 conversion ever occurs on fluid levels, so phase jumps $|\Delta S| \gg \pi\tilde{\hbar}$ pose no difficulty and the box-size limit of section 4.3 does not apply. The fluid equations

$$\partial_{\tilde{t}}\rho = -\nabla\cdot(\rho\nabla S), \qquad \partial_{\tilde{t}}S = -\tfrac{1}{2}|\nabla S|^2 - \tilde{\Phi}, \tag{4.3}$$

are integrated with upwind fluxes (continuity), the Sethian–Osher Hamiltonian (Hamilton–Jacobi) and a three-stage SSP Runge–Kutta scheme; the quantum-pressure term is deliberately omitted on fluid levels, which are restricted by the Madelung refinement criteria $C_1 = |\Delta_x\sqrt{\rho}|/\sqrt{\rho}$ and $C_2 = |\Delta_x S|/\tilde{\hbar}$ to regions where it is negligible. The fluid timestep obeys the advection condition $\Delta\tilde{t} \le C\,\Delta x/|\nabla S|_{\max}$ — linear in Δx, in contrast to the quadratic kinetic condition of explicit wave schemes.

At fluid–wave boundaries three couplings are required. (i) *Prolongation* interpolates (ρ, S) linearly and converts to $\psi = \sqrt{\rho}\,e^{\mathrm{i}S/\tilde{\hbar}}$ when the child is a wave level; interpolating the smooth (ρ, S) rather than the oscillatory ψ avoids creating interference artefacts. (ii) *Restriction* must conserve mass: plain complex averaging destroys it under interference ($|\langle\psi\rangle|^2 \le \langle|\psi|^2\rangle$), so wave children are restricted with amplitude $\sqrt{\langle|\psi|^2\rangle}$ and the phase of $\langle\psi\rangle$; wave-to-fluid restriction rebases the parent action by continuity. (iii) *Wave ghost cells* at coarse-fine faces are Dirichlet values interpolated from the parent (ρ, S) and converted to ψ; this is phase-coherent because the HJ equation evolves the same action as the Schrödinger phase up to the quantum-pressure correction. Since a Dirichlet boundary makes the Crank–Nicolson update non-unitary, we restore exact global conservation by flux bookkeeping: the discrete identity

$$\Delta|\psi_i|^2\big|_{\mathrm{face}} = -\frac{\tilde{\hbar}\,\Delta\tilde{t}}{\Delta x^2}\,\mathrm{Im}\big(\bar{\psi}_i^*\,\psi_{\mathrm{g}}\big)\,, \qquad \bar{\psi} \equiv \tfrac{1}{2}(\psi^n + \psi^{n+1}), \tag{4.4}$$

gives the exact mass exchanged through each ghost face, which is credited back to the adjacent parent cell (with an MPI reverse reduction for remote parents). In a deliberately hostile stress test ($15\,h^{-1}$Mpc, 256^3 base, $m_a = 10^{-23}$ eV, wave levels covering the collapsed regions), total mass drift at fixed epoch improves from 12.2 per cent (no coupling) to 0.12 per cent with conservative restriction and flux bookkeeping; fluid-only configurations conserve mass to machine precision.

4.5 Validation

Figure 1 shows the primary code test: power spectrum growth measured between snapshots of a $15\,h^{-1}$Mpc, 256^3 run with $m_a = 10^{-23}$ eV started at $z = 30$, compared with linear theory at the same epochs. Growth ratios relative to the initial snapshot are used throughout, which removes the overall normalisation of the initial conditions. The pure wave run (left) tracks linear theory (within $\sim$15 per cent, consistent with the onset of nonlinearity and the dark-matter/total growth difference) precisely until the velocity-Nyquist limit of section 4.3 is reached at $z \simeq 9$ for this configuration — beyond it the aliased phase freezes and then scrambles the growth, an empirical confirmation of the criterion. The HJM fluid run (right) of the identical configuration continues to grow through $z = 4$, with the late-time high-k excess over linear theory reflecting genuine nonlinear growth ($\Delta^2 \sim 1$).

The hybrid machinery was further exercised on a $128\,h^{-1}$Mpc, 1024^3 fluid-level run with $m_a = 10^{-22}$ eV — a configuration whose initial rms phase jump of 5.7π per cell excludes any pure wave treatment. With the persistent unwrapped action the measured growth between $z = 30$ and $z = 9$ is uniform in k and consistent with linear theory (whereas a wrapped-phase treatment stalls entirely), total mass is conserved to machine precision over hundreds of steps, and the advection-limited timestep exceeds the explicit wave-kinetic condition by two to three orders of magnitude at high redshift. Figure 2 shows the density field at $z = 3.5$. A full production hybrid configuration — wave levels confined to collapsed regions by the C_1/C_2 criteria, with the associated refinement-cost controls — together with FDM-specific science validation (soliton cores, interference statistics, suppressed small-scale power against Hu et al. [15]-type transfer functions) is deferred to a companion paper.

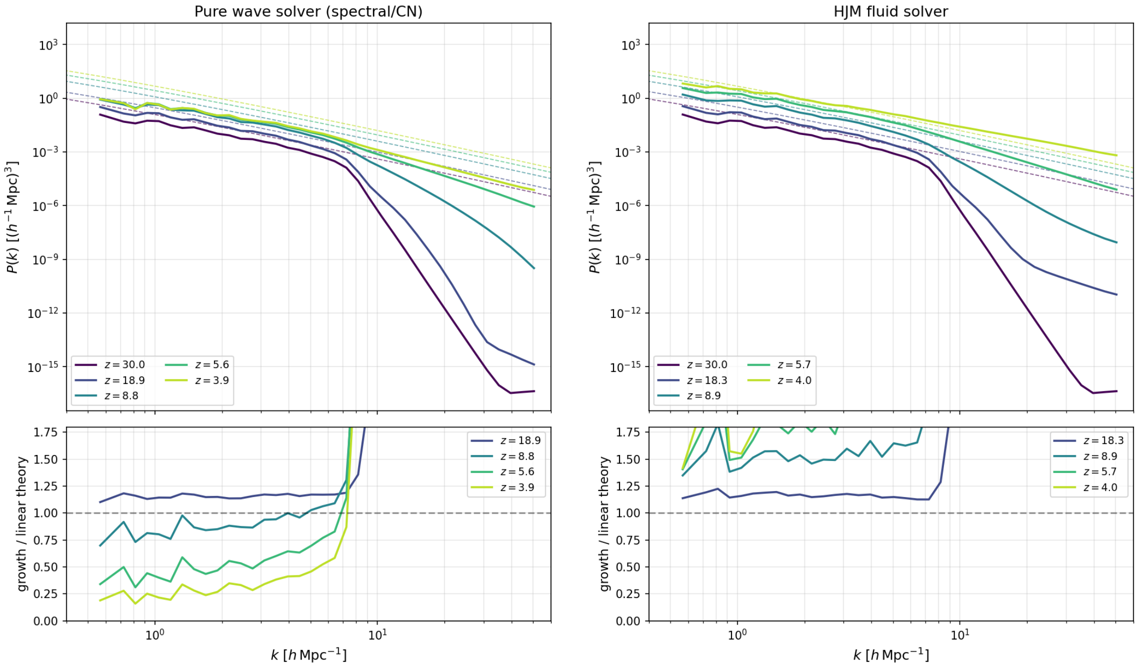

Figure 1. Growth validation on a $15\,h^{-1}$Mpc, 256^3 grid with $m_a = 10^{-23}$ eV. Solid curves: measured $|\psi|^2$ power spectra; dashed: linear theory; bottom panels: measured growth relative to the $z = 30$ snapshot divided by the linear-theory growth. *Left*: pure wave solver — growth follows linear theory until the velocity-Nyquist limit ($z \simeq 9$ here, section 4.3) and freezes beyond it. *Right*: HJM fluid solver on the same configuration — growth continues through $z = 4$; the high-k excess at late times is nonlinear growth.

5 Evaporating Primordial Black Holes

The PBH module models dark matter that is partly composed of Hawking-evaporating primordial black holes [8, 13]. Rather than tracking per-particle masses, it uses one global mass function: all DM particles share the mixed-mass factor

$$w(a) \;=\; 1 - f_{\rm PBH} + f_{\rm PBH}\, g(a), \tag{5.1}$$

where $g(a)$ is the surviving PBH mass fraction from a precomputed evaporation table and $f_{\rm PBH}$ (`pbh_fraction`) the initial PBH share of the dark matter. Particle masses are rescaled exactly by table arithmetic — there is no per-particle state and no evaporation CFL constraint. The energy released between steps follows the cumulative heating kernel $\tilde{Q}(a) = \int q\, g\, \mathrm{d}t$; it is either deposited as local heat in the gas (`pbh_energy_sink='local_heat'`, optionally amplified by a linear-response boost factor) or removed from the simulation (`'removed'`, appropriate for free-streaming decay products). Restart provenance (table checksum, normalisation epoch) is persisted so that a run cannot silently continue with a different evaporation model. The table generator supports arbitrary monochromatic or extended PBH mass functions.

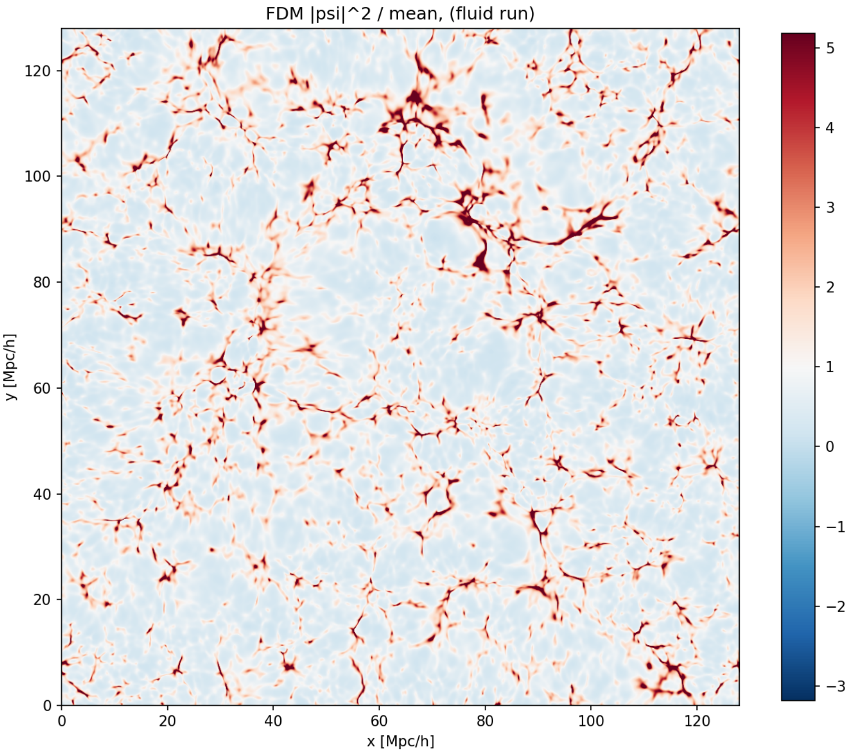

Figure 2. Dark-matter density slice of the $128\,h^{-1}$Mpc, 1024^3 hybrid-fluid FDM run ($m_a = 10^{-22}$ eV) at $z = 3.5$, a regime inaccessible to pure wave discretisations at this resolution (section 4.3).

6 Conclusions

We have assembled, in one AMR code, the four families of non-standard cosmological physics most often invoked beyond ΛCDM — self-interacting dark matter, a comprehensive dark-energy and modified-gravity battery, fuzzy dark matter, and evaporating primordial black holes — behind uniform namelist switches and on shared solver infrastructure. Beyond the implementations themselves, we highlight the verification methodology: every model is checked at the equation level against its original reference, at the solver level against analytic linear responses, and end-to-end against an initial-conditions-consistent forward model, with per-effect isolation runs where a model bundles several couplings. We commend this three-level protocol to implementers of non-standard physics in any simulation code: in our experience each level catches defects invisible to the others, ranging from wrong-sign perturbation closures to factor-of-two integrator bookkeeping and MPI collective mismatches that only manifest on sparsely refined levels.

All modules recover ΛCDM bit-wise when disabled. The namelist interfaces, verification scripts and the simulation-suite tooling are distributed with the code.

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Data Availability

The code, namelist examples, verification drivers and the 64^3 test-suite configuration are available from the CURAMSES repository. Simulation outputs will be shared on reasonable request.

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